WEEK 2 · 16–19 NOVEMBER 2026

Feature extraction and dimensionality reduction

From 352 redundant bands to 16 features you can defend

Week 2 · Session 2 · practical: 13_feature_extraction_dim_reduction.ipynb

LEARNING OBJECTIVES

What you will be able to do

- Explain why neighbouring hyperspectral bands are redundant, and what the Hughes phenomenon costs you
- Apply **PCA** and **MNF**, and say which one to use when noise matters
- Select a small set of *physical* bands, and recognise when a supervised selector is fooled by artefacts
- Compute **derivative** features and the red-edge position
- Choose narrow-band indices and absorption depths that Sentinel-2 cannot reproduce
- Assemble a compact, interpretable feature stack for classification

1. Why reduce at all?

Redundancy, and what it does to a classifier

THE STARTING POINT

A Tanager pixel is 352 numbers.
It is not 352 pieces of information.

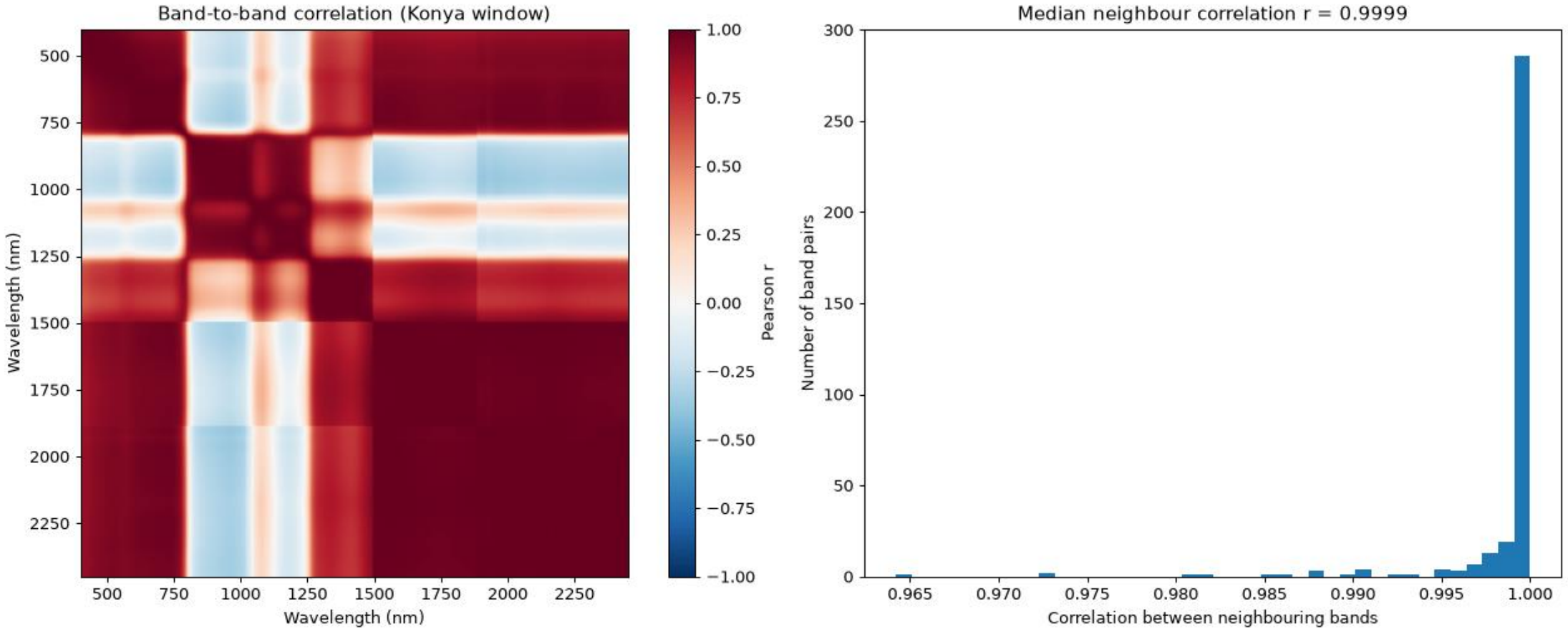
Neighbouring bands are 5 nm apart, but real surface spectra change over tens of nanometres. In the Konya window, three principal components already carry 99 % of the variance.

Redundancy: the bands repeat each other

- Tanager samples every ~5 nm; most reflectance spectra vary smoothly over tens of nanometres
- Almost every pair of **neighbouring bands has r > 0.99**
- The correlation matrix shows **blocks**: visible and SWIR behave alike, the NIR plateau opposes them
- The flip happens at the **red edge** — vegetation is dark in the visible, bright in the NIR
- **Effective dimensionality** of a scene is far below the band count

KONYA, TÜRKIYE

The correlation matrix of 352 bands



Band-to-band correlation for a 200 × 200 px window over irrigated fields, Konya

The Hughes phenomenon (Hughes, 1968)

- With a **fixed** number of training samples, accuracy rises with features, peaks, then **falls**
- The model has more parameters than the data can constrain reliably
- A Gaussian classifier needs a covariance matrix of $d(d+1)/2$ numbers per class
- For d = 352 bands that is **62,128 parameters per class** — from a few thousand labelled pixels
- Still true in the deep-learning era: it is why we pretrain, regularise and reduce

Two families of solutions

	Feature extraction	Feature selection
What it does	Builds new features from all bands	Keeps a subset of real bands
Examples	PCA, MNF, derivatives, indices	Variance/correlation, mutual information
Interpretable?	Mixtures of all bands — harder	Yes, each band has a wavelength
Good for	Compression, denoising, classifiers	Explaining, cheap sensors, fast products

2. Principal Component Analysis

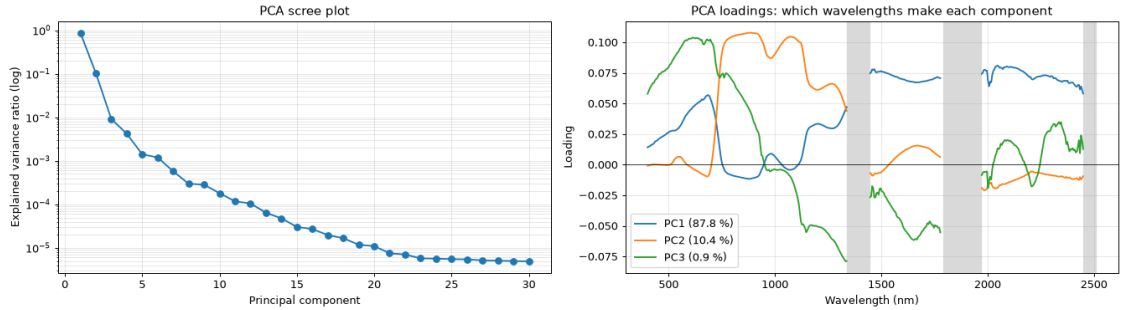
Ordering by variance

What PCA actually computes

- Rotates the data onto orthogonal axes ordered by **variance explained**
- The axes are the eigenvectors of the band covariance matrix: **$\Sigma \cdot v = \lambda \cdot v$**
- $PC_k = (x - \bar{x})^T \cdot v_k$ — a weighted sum of all bands
- The **loadings** v_k tell you which wavelengths drive that component
- Covariance PCA is dominated by bright bands; **standardising first** gives every band equal weight
  - Always state which of the two you used — the components differ

Konya: three components hold 99 % of the variance

- Explained variance: **87.8 %**, 10.4 %, 0.9 % for PC1–PC3
- 3 components reach 99 %, 10 components reach 99.9 %
- PC1 is essentially **brightness**
- PC2 is a **vegetation** axis, driven by the NIR plateau
- PC3 contrasts visible/NIR against SWIR — moisture and dry matter



Explained variance per component

THE LIMITATION THAT MATTERS

PCA ranks by variance.
Variance is not the same as information.

Subtle diagnostic features — the xanthophyll signal behind PRI, a weak clay absorption — have low variance. They land in high-order components, mixed with noise. MNF fixes the noise half of that problem.

3. Minimum Noise Fraction

Ordering by signal-to-noise

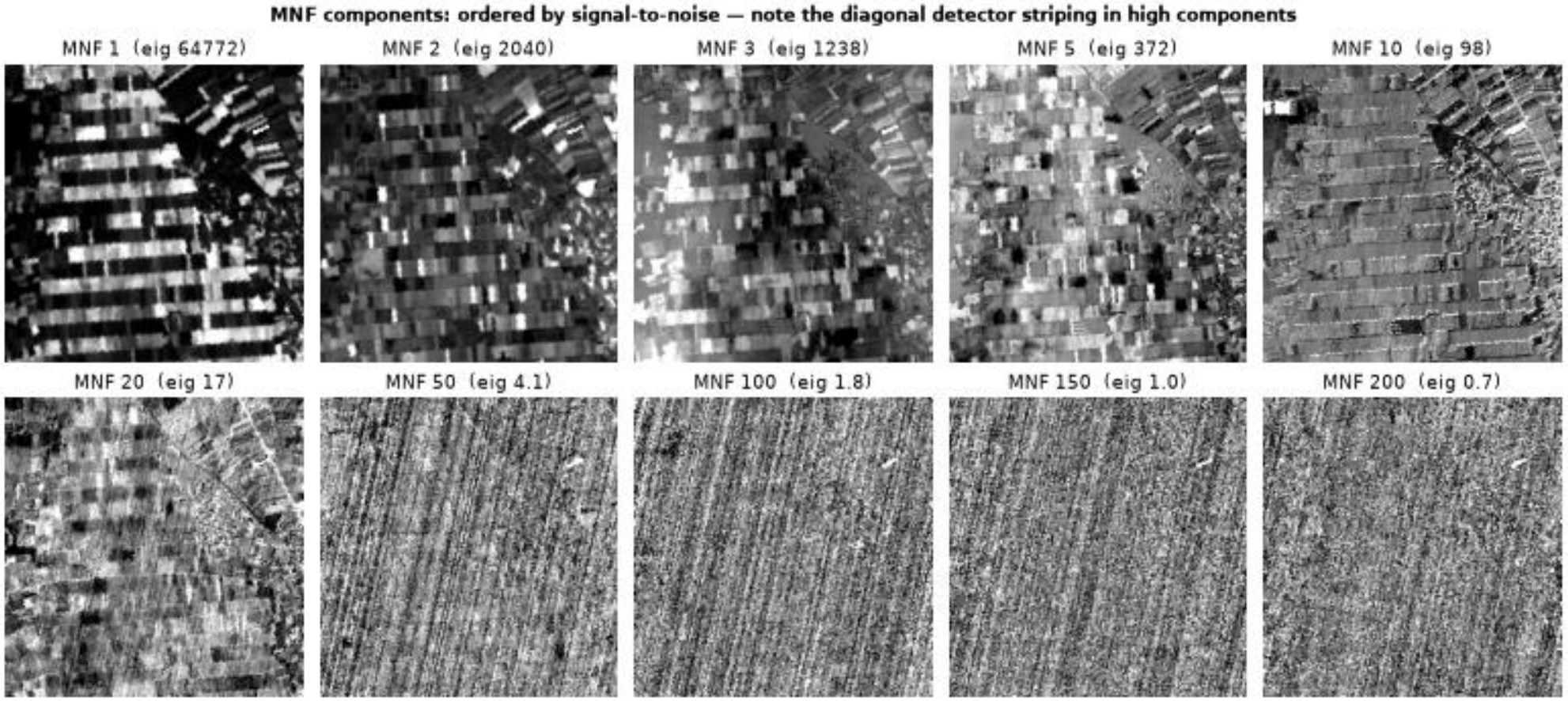
MNF = PCA after whitening the noise (Green et al., 1988)

- **1. Estimate the noise covariance** Σ_N — the classic *shift difference* trick
  - In a homogeneous area, x(i,j) − x(i,j+1) is almost pure noise, with covariance 2Σ_N
- **2. Whiten**: transform with W = V_N·Λ_N^(−1/2) so noise has unit variance in every direction
- **3. PCA on the whitened data** — components are now ordered by **SNR**
- Eigenvalue ≈ SNR + 1, so components with eigenvalue ≈ 1 are pure noise

Estimating the noise honestly

- The notebook searches the scene for the **most homogeneous 40 × 40 px window** (lowest CV at 1650 nm)
- Shift-difference and the file's own uncertainty layer agree in the blue, differ by 1.5–3.5× in NIR/SWIR
- Real texture is counted as noise, so shift difference is an **upper bound**
- Noise is of order **0.002–0.007 reflectance** — small against signals of 0.05–0.5, not negligible
- Tanager's noise covariance is **near-singular** (eigenvalues ~1e−15)
  - Add a small **ridge** — 0.1 % of mean noise variance — before whitening; standard practice

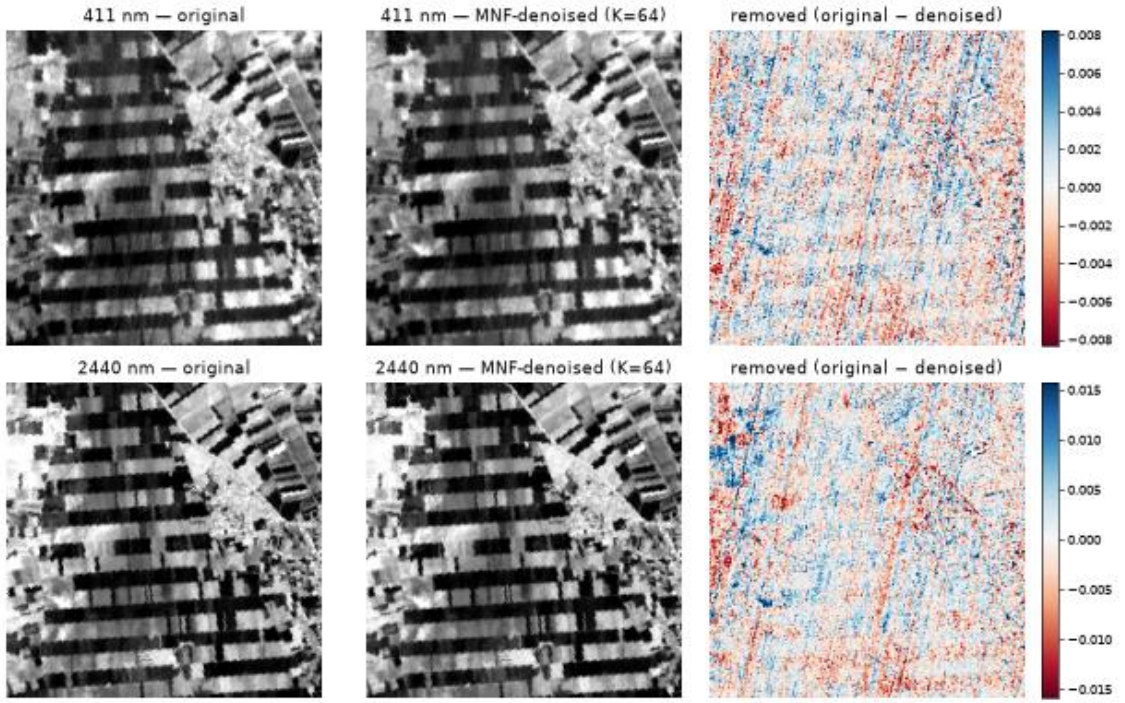
MNF isolates sensor artefacts



MNF components: low orders show fields and the town, high orders show detector striping

Denoising with the inverse transform

- Keep the first K components, zero the rest, transform back
- K = 64 of 352 gave a median change of **0.0029 reflectance**
- Biggest gains at the noisy ends: 410 nm and the 2440 nm SWIR edge
- **Rule of thumb:** use MNF, not PCA, as classifier input when noise matters
- Always inspect the *removed* image — it must look like stripes and speckle



Original vs MNF-denoised spectrum

4. Band selection

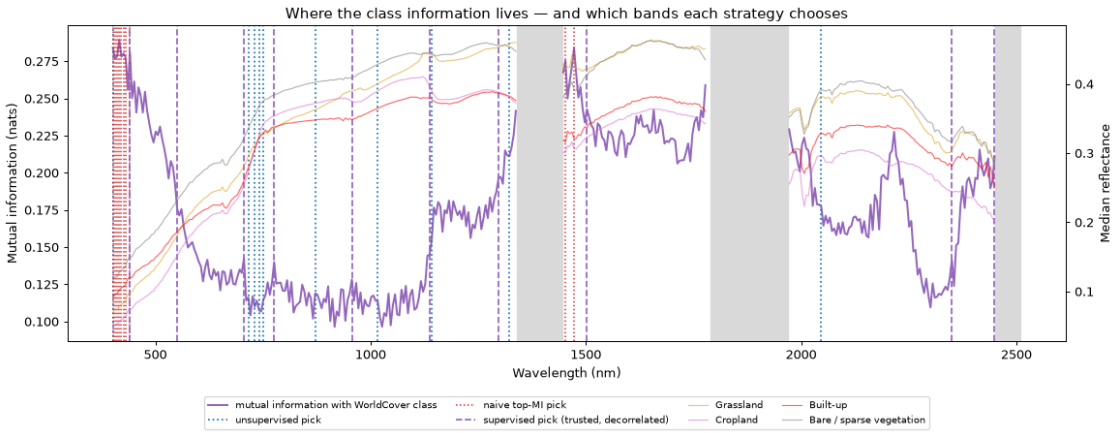
Keeping bands you can name

Two strategies

- **Unsupervised** — pick bands with high variance that are least correlated with those already chosen
- **Supervised** — rank bands by **mutual information** with class labels, then remove redundancy
  - In the spirit of mRMR: maximum relevance, minimum redundancy (Peng et al., 2005)
- Labels here: ESA WorldCover 2021, pure 30 m pixels only
- The labels are **four years older** than the image — treat them as approximate

A supervised selector can be fooled

- Naive top-MI picks cluster at **400–430 nm** and **1450–1480 nm**
- Both are low-SNR or atmosphere-affected regions
- **Noise hides redundancy** — noisy bands look independent of each other
- **Spurious correlation** — haze and water vapour vary with terrain, and terrain correlates with land cover
- It predicts the label for the wrong reason, and will not transfer



Bands chosen by each strategy

The fix: constrain the search

- Consider only **trustworthy bands** — ≥ 430 nm, away from the water-vapour shoulders
- After choosing a band, **exclude everything correlated above |r| = 0.95**
- This forces the picks to spread across the spectrum
- Result: 441, 551, 706, 776, 957, 1137, 1297, 1503, 2350, 2449 nm
  - Physically sensible: blue, green, red edge, NIR, water bands, SWIR minerals

Ten features can match 352

Feature set	Features	Accuracy
All bands	352	0.870
10 naive top-MI bands	10	0.695
10 MI bands, constrained	10	0.869
10 unsupervised bands	10	0.835
10 evenly spaced bands	10	0.849
10 PCA components	10	0.900
10 MNF components	10	0.859

5. Shape features

Derivatives, the red edge and narrow-band indices

Derivatives emphasise shape, not brightness

- Suppress additive offsets: soil background, residual haze
- Sharpen overlapping features
- They **amplify noise** — always smooth first (Savitzky–Golay, 1964)
- The best-known derivative feature is the **red-edge position (REP)**
- REP shifts to **longer** wavelengths as chlorophyll increases; shorter under stress

Red-edge position, four-point method (Guyot & Baret, 1988)

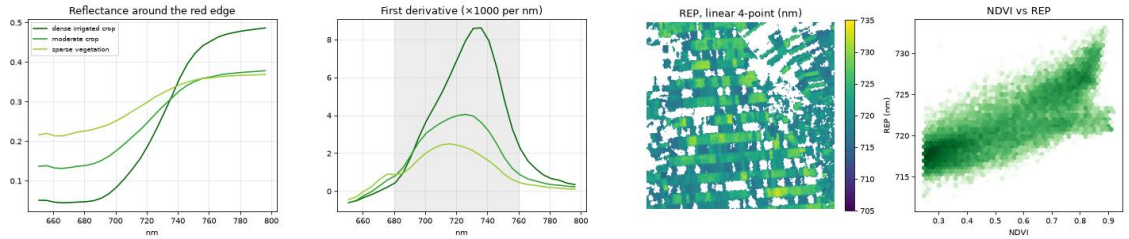
```
# R_re is the midpoint between the red trough and the NIR shoulder
R_re = (R670 + R780) / 2

# linear interpolation between the 700 and 740 nm bands
REP  = 700 + 40 * (R_re - R700) / (R740 - R700)
```

Four bands, no fitting, no smoothing — and it agrees with the max-derivative REP at r = 0.87

REP keeps working after NDVI saturates

- REP over vegetated pixels: 716.4 / 720.3 / 727.9 nm (5th / 50th / 95th)
- Between NDVI 0.75 and 0.9, REP still spans **more than 10 nm**
- NDVI saturates; the red edge keeps separating dense canopies
- The two branches at high NDVI are different crops or growth stages
- This is why red-edge features dominate nitrogen and stress monitoring



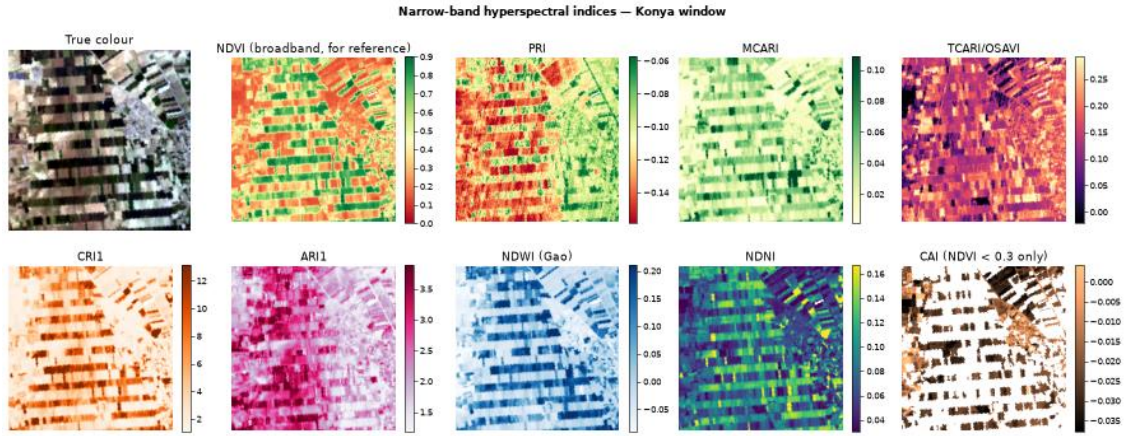
Derivative spectra and the NDVI–REP relationship

Narrow-band indices: what only a spectrum can give

Index	Bands (nm)	Sensitive to	From S-2?
PRI	531, 570	Xanthophyll cycle, light-use efficiency	No
MCARI	550, 670, 700	Chlorophyll, reduced soil effect	Approx.
TCARI/OSAVI	550, 670, 700, 800	Chlorophyll, ignores canopy density	Approx.
CRI1	510, 550	Carotenoids	No
ARI1	550, 700	Anthocyanins (senescence, stress)	Approx.
NDWI (Gao, 1996)	860, 1240	Canopy liquid water	No
NDNI	1510, 1680	Canopy nitrogen	No
CAI	2031, 2101, 2211	Cellulose and lignin: crop residue	No

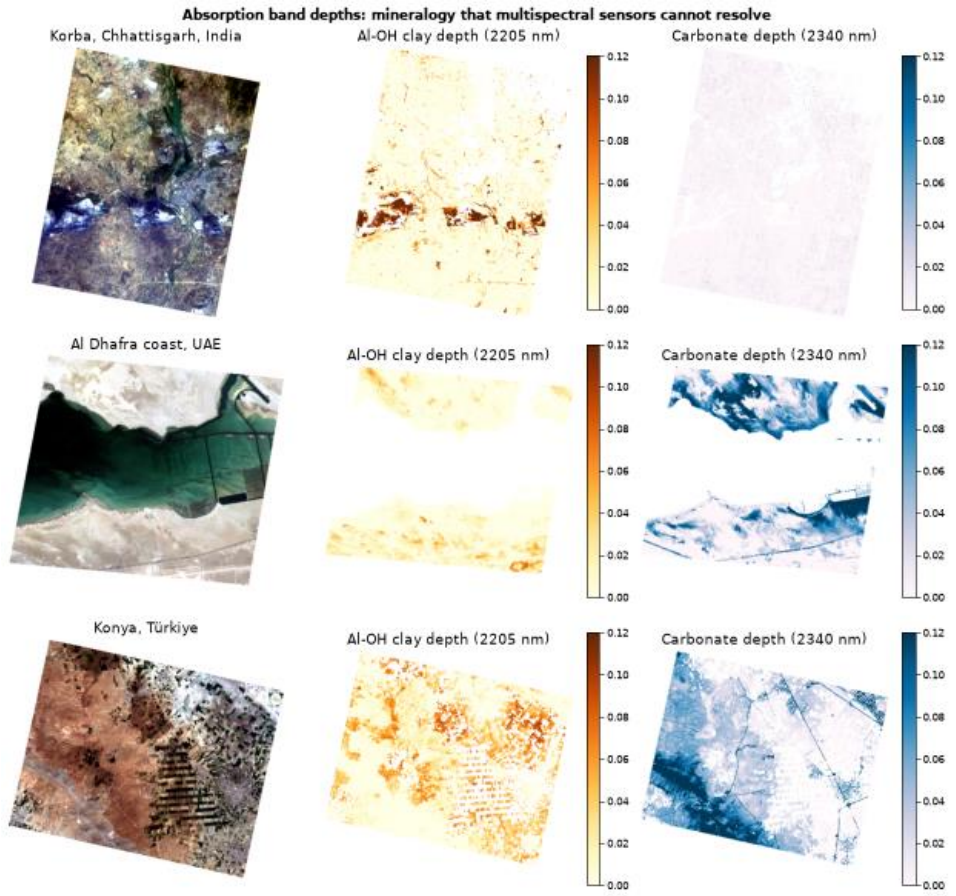
Reading the index maps

- **CAI** is only meaningful on **non-green** pixels — the notebook masks NDVI > 0.3
  - In green canopies, leaf water dominates the 2000–2200 nm shape
- Among bare fields, higher CAI marks **dry residue and stubble**
- **NDWI (Gao)** peaks on the irrigated strips
- **PRI** is a physiological signal and is noisy from space — compare within one scene only



Narrow-band indices over the Konya window

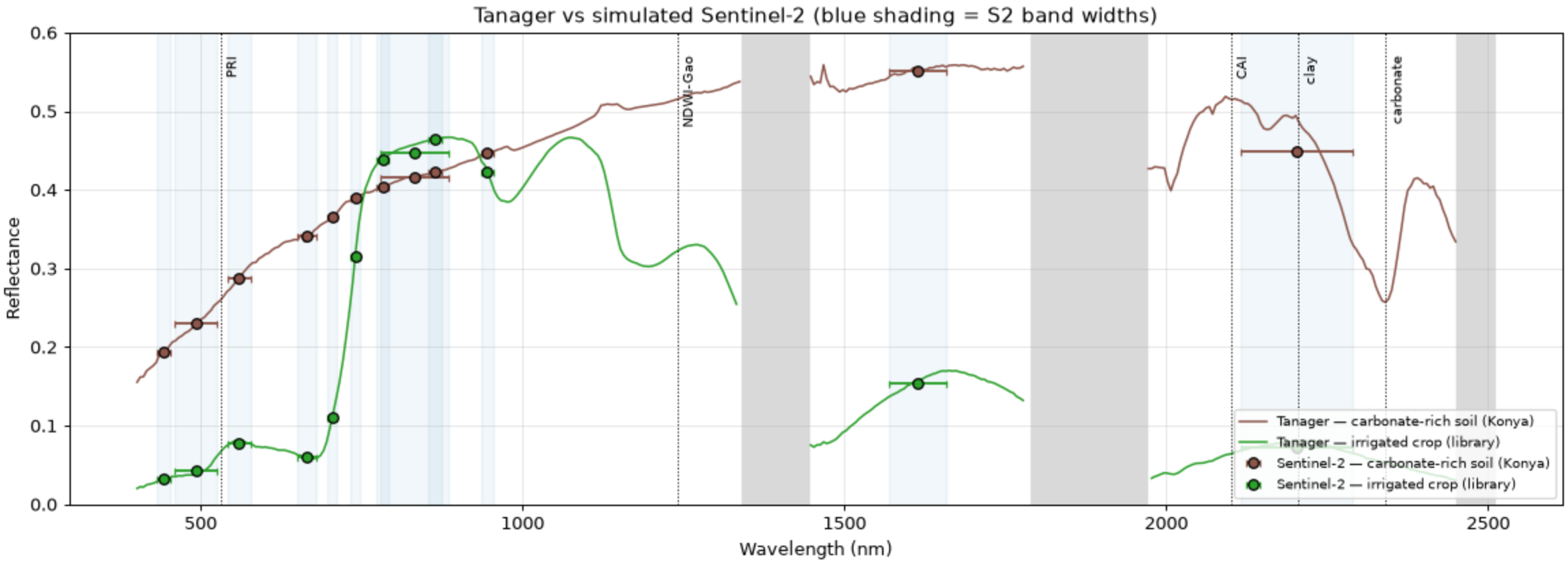
Absorption band depths: geology in the SWIR



Clay (2205 nm) and carbonate (2340 nm) depths, three scenes

- Depth **D = 1 – R_centre / R_continuum**, continuum = line between two shoulders
- **Al–OH in clays** near 2200 nm — kaolinite doublet at ~2165 and ~2205 nm
- **Carbonates**: calcite ≈ 2340 nm, dolomite ≈ 2320 nm
- Mask vegetation and water — they have their own SWIR shapes
- Korba clay on mine overburden · Al Dhafra and Konya carbonate

The same spectra through Sentinel-2's bands



Konya carbonate soil and irrigated crop, resampled to Sentinel-2 bands

6. The feature stack

Sixteen numbers instead of 352

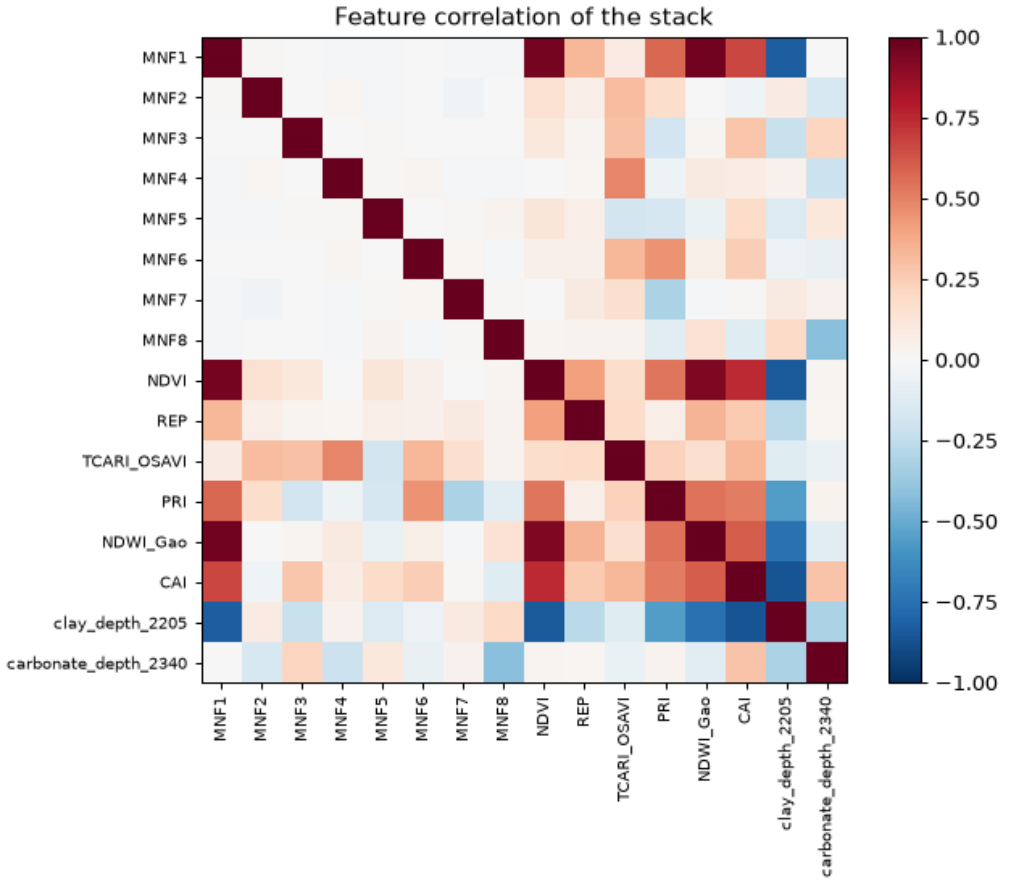
16 FEATURES

A stack you can defend to a domain expert

Group	Features
Extracted, noise-aware	MNF 1–8
Vegetation physiology	NDVI, REP, TCARI/OSAVI, PRI, NDWI (Gao)
Dry matter and soil	CAI, clay depth, carbonate depth

Still some redundancy — and that is fine

- MNF components are uncorrelated by construction
- MNF1, NDVI, NDWI (Gao) and CAI all respond to green vegetation
- Clay depth is **negatively** correlated with them — canopies hide the soil feature
- For a classifier, correlated features are acceptable
- For **interpretation**, pick one representative per group



Correlation between the 16 stacked features

PRACTICAL

Over to you

▶ `13_feature_extraction_dim_reduction.ipynb` · `120 min`

- Run PCA and MNF on the Konya window and compare the component images
- Reproduce the naive vs constrained band selection and the random-forest comparison
- Map REP and compare it with NDVI on dense canopies
- Compute CAI, clay and carbonate depths, then resample to Sentinel-2 and see what survives
- Build and save the 16-feature stack for tomorrow's classification session

KEY TAKEAWAYS

Recap

- Bands are **redundant** (neighbour r > 0.99); too many features hurt — **Hughes, 1968**
- **PCA** orders by variance; **MNF** orders by SNR, isolates striping and enables denoising
- Supervised band selection can lock onto **artefacts** — constrain it and check the wavelengths
- **Ten well-chosen features matched all 352** on this scene
- **REP** keeps discriminating after NDVI saturates
- **Narrow-band indices and band depths** are the hyperspectral payoff — most are impossible on S-2
- For your project: a small, **justified** stack beats all 352 bands

Next: Session 3 — Material discrimination and target detection

References and further reading

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